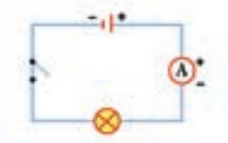
- ❖ Parallel Circuit — A circuit in which there are multiple paths for electricity to flow. Each load connected in a separate path receives the full circuit voltage, and the total circuit current is equal to the sum of the individual branch currents.
- ❖ Series Circuit — A circuit in which there is only one path for electricity to flow. All of the current in the circuit must flow through all of the loads.
- ❖ Short Circuit — When one part of an electric circuit comes in contact with another part of the same circuit, diverting the flow of current from its desired path.
- ❖ One unit of coulomb is charge of approximately 6.242×10^{18} protons or electrons.
- ❖ The potential difference between any two points is the amount of energy needed to move one unit of electric charge from one point to the other.
- ❖ Electrical conductivity or specific conductance is the measures a material's ability to conduct an electric current
- ❖ Electrical resistivity is the property of a material that quantifies how strongly that material opposes the flow of electric current.
- ❖ The sources which produce the small amount of electricity for shorter periods of time is called as electric cell or electro chemical cells.
- ❖ Electrolytes : A substance that dissociates into ions in solution and acquires the capacity to conduct electricity. Sodium, potassium, chloride, calcium, and phosphate are examples of electrolytes.

EVALUATION

I. Choose the correct answers

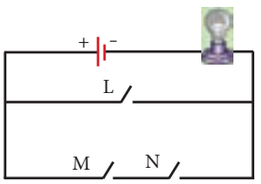
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1. In the circuit diagram below, 10 units of electric charge move past point x every second What is the current in the circuit ----------



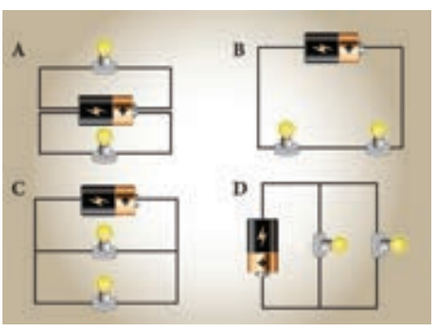
a) 10 A b) 1 A
c) 10 V d) 1 V

2. In the circuit shown, which switches (L,M or N) must be closed to light up the bulb?



a) switch L only
b) switch M only
c) Switch M and N only
d) either switch L or switches M and N

3. Small amounts of electrical current are measured in milliampere (mA). How many milliampere are there in 0.25 A ?
a) 2.5 mA b) 25 mA
c) 250 mA d) 2500 mA
4. In which of the following circuits are the bulb connected in series?



II. Fill in the blanks.

1. The direction of conventional current is ---------- to electron flow.

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